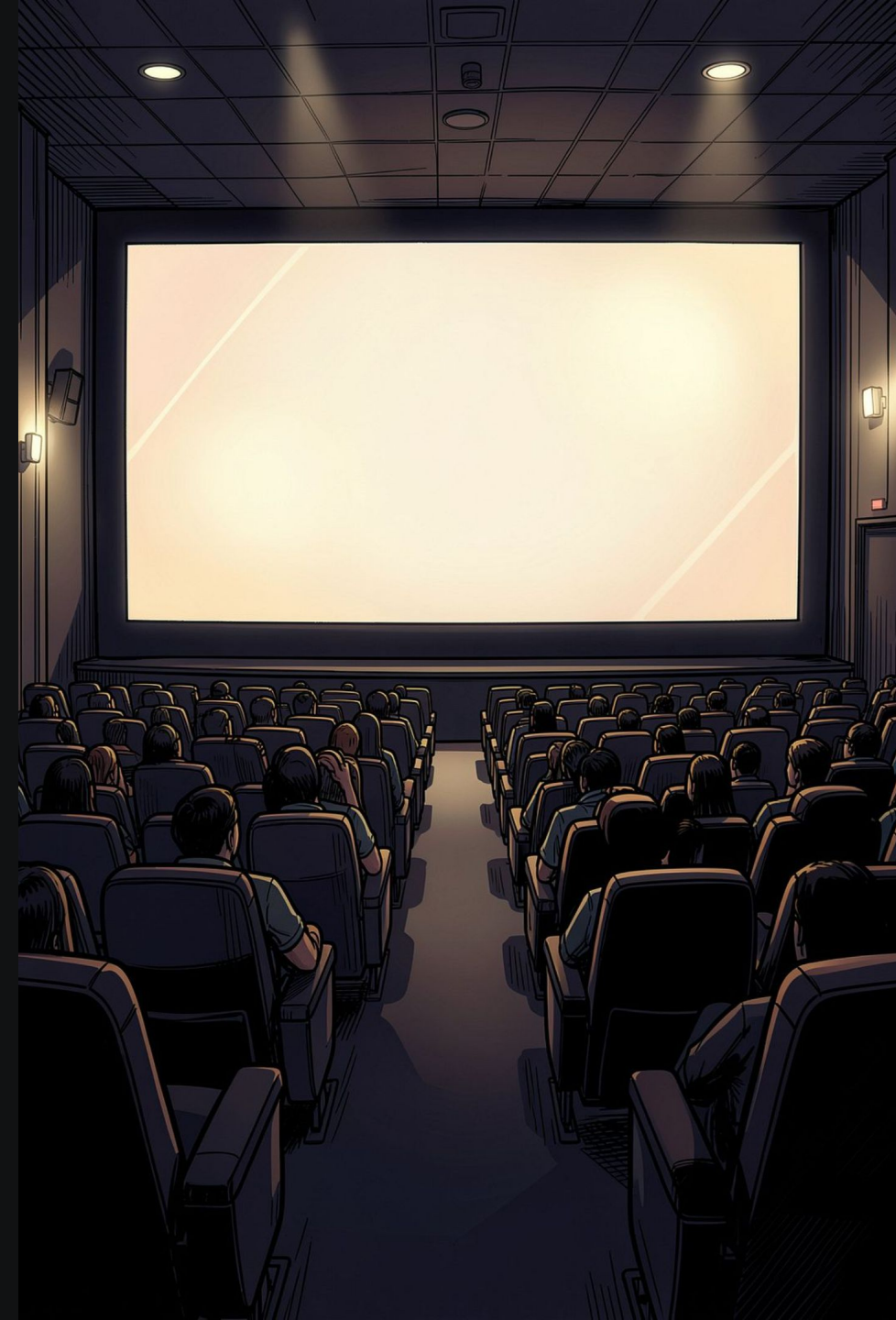


Movie Recommendation System

Internship Report · B.Tech CSE (Core) · IILM University, Greater Noida

Student: Harsh Vardhan | Roll No: 25SCS1003003013 | 2026–27



Declaration & Acknowledgement

Candidate's Declaration

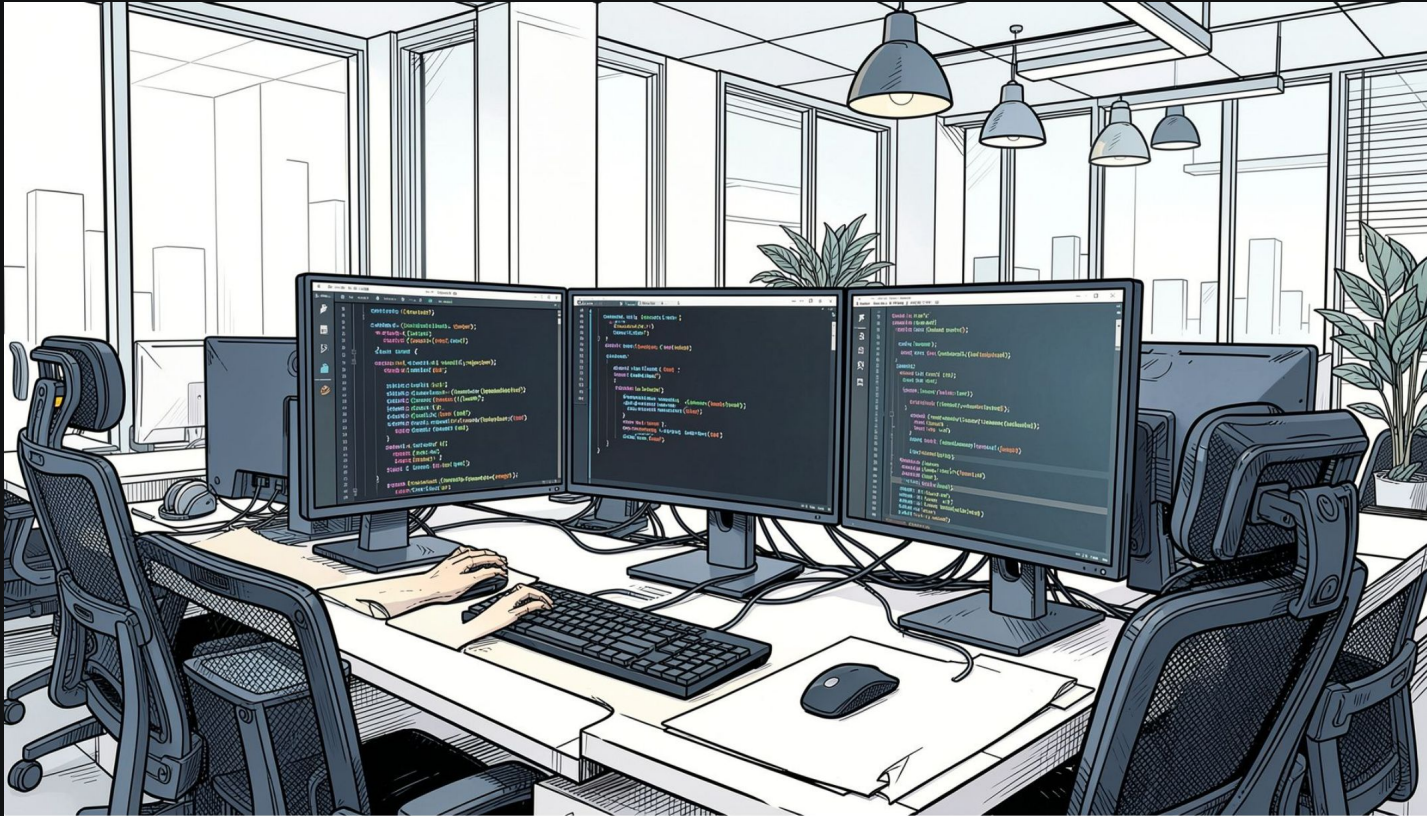
I Harsh Vardhan, declare this report is my original work, submitted solely to IILM University for the B.Tech CSE degree. All references are cited; no material is copied from other sources.

Acknowledgement

Gratitude to all who supported this internship — for their guidance, constructive criticism, and advice. Special thanks to my parents for their blessings.



Internship Offer — Codec Technologies



1-MONTH INTERNSHIP

Artificial Intelligence Intern

Codec Technologies is a global IT and business consultancy platform operating across **27+ countries**, empowering innovation and strategic growth.

- **Duration:** 05/08/2026 – 06/09/2026
- **Mode:** Pan India, Hybrid
- **Reporting to:** Assigned Project Head(s)
- **Issued by:** Dr. Anurag Shrivastava, Talent Acquisition Manager

❏ Exceptional interns may qualify for a scholarship based on weekly performance reviews.



The Challenge

Key Challenges

Made with **GAMMA**

Project Objectives

01

Understand fundamentals

Study movie recommendation systems and their real-world applications.

02

Collect & clean data

Gather and prepare movie metadata for reliable processing.

03

Engineer features

Build meaningful representations from genres, keywords, cast, crew, and descriptions.

04

Vectorize & model

Apply TF-IDF vectorization, train and compare similarity-based models.

05

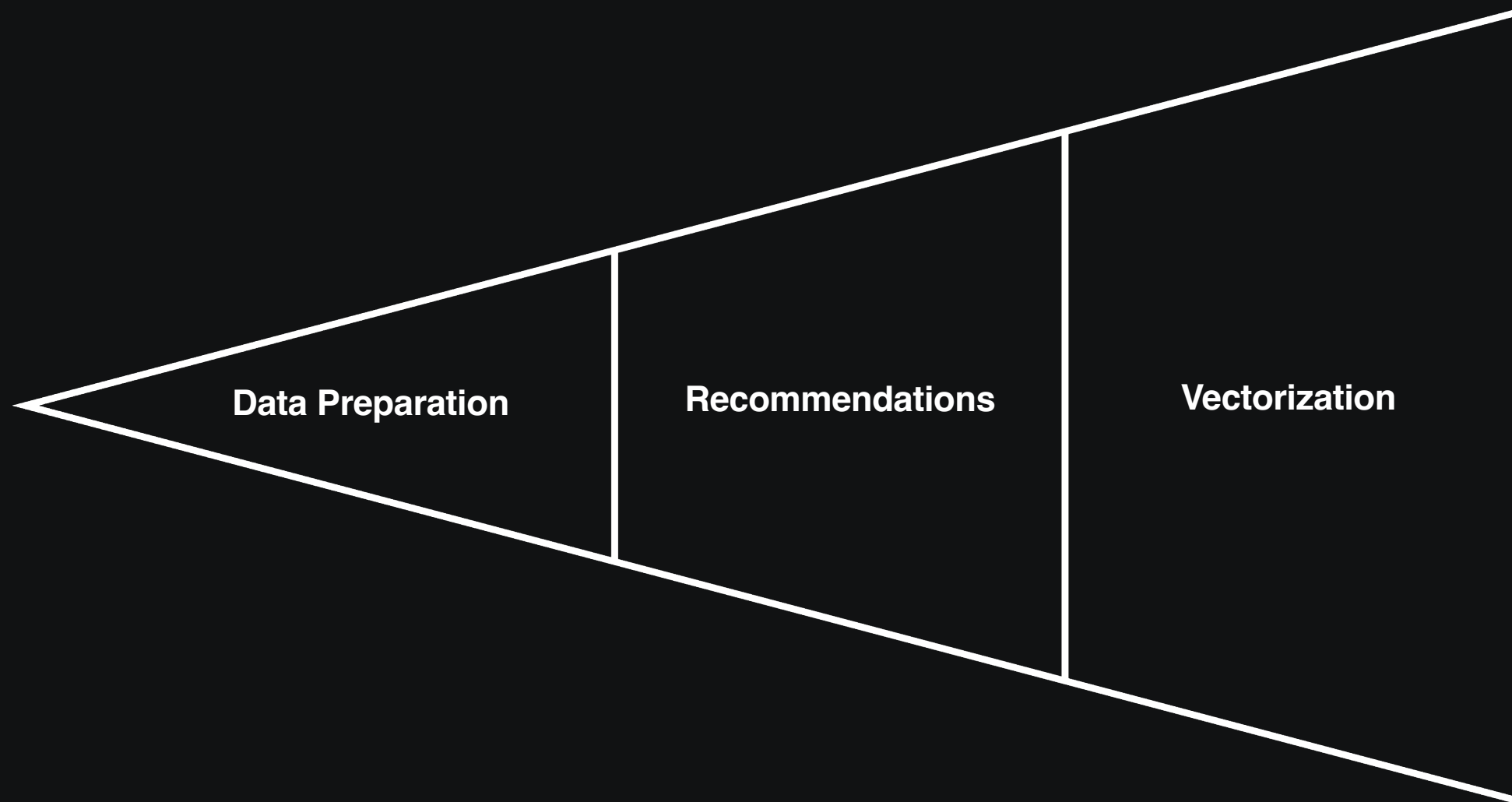
Improve & rank

Refine features, tune parameters, explore ensemble modeling, and generate ranked recommendations.

Technologies & Tools

Python	Primary language for data processing and model development
Pandas & NumPy	Data loading, cleaning, filtering, and numerical computation
Scikit-learn	TF-IDF vectorization, cosine similarity, and ML utilities
NLTK	Tokenization and text normalization
Matplotlib	Exploratory charts and data visualization
Jupyter / VS Code	Development, experimentation, and documentation
Git	Version control for tracking project changes

System Architecture



The modular design allows each component to be tested and improved independently. TF-IDF can later be replaced by word embeddings or transformer-based representations without redesigning the entire pipeline.

Core ML Methodology

TF-IDF Vectorization

Term Frequency–Inverse Document Frequency increases the weight of distinctive terms in a movie's description while reducing the influence of common words.

$$\text{TF-IDF}(t, d) = \text{TF}(t, d) \times \text{IDF}(t)$$

Cosine Similarity

Compares two movie vectors — a higher score indicates more similar content.

$$\text{Cosine}(A, B) = \frac{A \cdot B}{\|A\| \|B\|}$$

Ensemble Modeling

Multiple similarity signals are combined with tunable weights for balanced recommendations:

$$\text{Score} = w_1(\text{Genre}) + w_2(\text{Keyword}) + w_3(\text{Overview}) + w_4(\text{Cast/Crew})$$

- Genre similarity controls category compatibility
- Keyword similarity captures themes
- Overview similarity captures plot relationships
- Cast/Crew similarity adds creator information

Scope & Expected Outcomes

In Scope

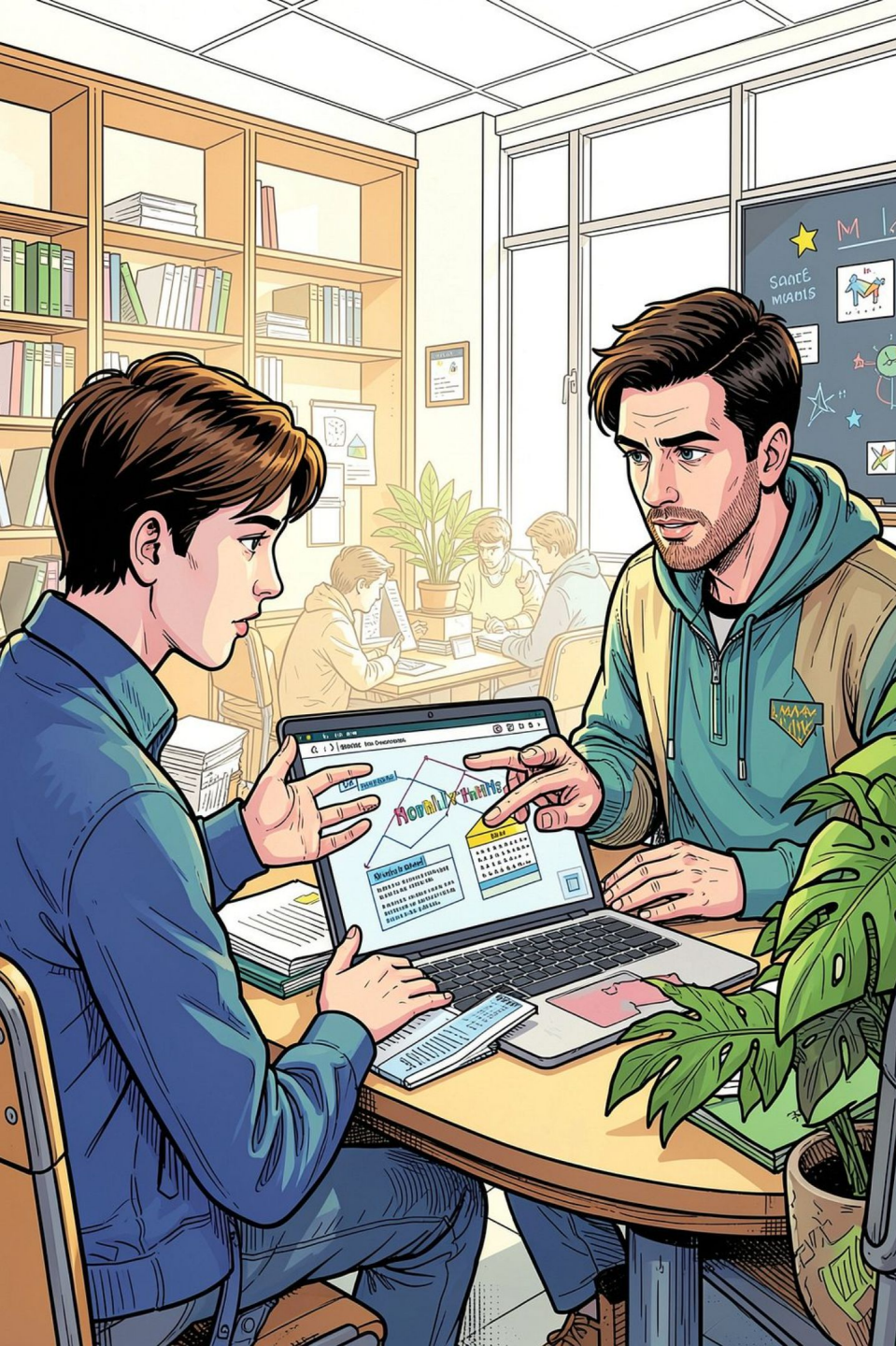
- Movie metadata preprocessing and validation
- Text-based feature construction from descriptive fields
- TF-IDF vectorization and cosine similarity
- Model training, comparison, and ranking
- Basic evaluation and qualitative inspection

Out of Scope / Future Extensions

- Real-time personalization using large-scale user histories
- Production-grade distributed serving and cloud deployment
- Full user authentication, payments, and streaming
- Commercial licensing of copyrighted movie data



Future path: hybrid recommendation combining content features with user ratings, collaborative filtering, and deep learning embeddings.



Conclusion

This project demonstrates how **ML and NLP** can reduce information overload in digital entertainment — from data cleaning and feature engineering to TF-IDF vectorization and similarity-based ranking.

Academic Learning

Practical exposure to the full ML lifecycle: problem definition, data preparation, modeling, evaluation, and documentation.

Technical Foundation

Hands-on experience with Python, Pandas, Scikit-learn, NLP, TF-IDF, and ensemble modeling for recommender systems.

Future Path

Extendable to hybrid systems with collaborative filtering, semantic embeddings, and deep learning.